



Module 5

Google Cloud Platform and Services

Types of Cloud Services

Public cloud providers such as Google, Amazon, and Microsoft offer a range of services for deploying computing, storage, networking, and other infrastructures to run a wide array of business services and applications.

- Some cloud users are new companies that start in the cloud. They have never owned their own hardware and software.
- A company that starts on the cloud can choose services that best fit its application and architectural needs without having to consider existing infrastructure.
- Public cloud providers offer services that fall into four broad categories:
 - Compute resources
 - Storage
 - Networking
 - Specialized services such as Machine Learning Services

Cloud customers typically make use of services in more than one of these categories

Compute Resources

Computing resources come in a variety of forms in public clouds:

- ▣ **“Virtual Machines”** are a basic unit of computing resources and a good starting point for experimenting with the cloud. After you create an account with a cloud provider and provide billing information, you can use a portal or command-line tools to create VMs.
- ▣ **“Managed Kubernetes Clusters”** Google Cloud Platform gives you all the tools you need to create and manage clusters of servers. Many cloud users would rather focus on their applications and not the tasks needed to keep a cluster of servers up and running. In a managed cluster, you can specify the number of servers you would like to run and the containers that should run on them. You can also specify autoscaling parameters to optimize the number of containers running.

Compute Resources

Computing resources come in a variety of forms in public clouds:

- **“Serverless Computing”** Both VMs and managed kubernetes clusters require some level of effort to configure and administer computing resources.
- Serverless computing is an approach that allows developers and application administrators to run their code in a computing environment that does not require setting up VMs or kubernetes clusters. Google Cloud Platform has two serverless computing options: App Engine and Cloud Functions.

Storage

Public clouds offer a few types of storage services that are useful for a wide range of application requirements. These types include the following:

- **“Object Storage”** Object storage is a system that manages the use of storage in terms of objects or blobs. Objects are grouped into buckets. Each object is individually addressable, usually by a URL. Advantage of object storage is that it is serverless, There is no need to create VMs and attach storage to them.
- **“File Storage”** File storage services provide a hierarchical storage system for files. File systems storage provides network shared file systems. Google Cloud Platform has a file storage service called Cloud Filestore, which is based on the Network File System (NFS) storage system. File storage is suitable for applications that require operating system–like file access to files.



Storage

Public clouds offer a few types of storage services that are useful for a wide range of application requirements. These types include the following:

- **“Block Storage”** Block storage uses a fixed-size data structure called a *block* to organize data. Block storage is commonly used in ephemeral and persistent disks attached to VMs. With a block storage system, you can install file systems on top of the block storage, or you can run applications that access blocks directly.
- **“Caches”** Caches are in-memory data stores that maintain fast access to data. The time it takes to retrieve data is called latency. The latency of in-memory stores is designed to be submillisecond.

Networking

- When working in the cloud, you'll need to work with networking between your cloud resources and possibly with your on-premise systems.
- When you have multiple VMs running in your cloud environment, you will likely need to manage IP addresses at some point. Each network-accessible device or service in your environment will need an IP address.
- Internal addresses are accessible only to services in your internal GCP network. Your internal GCP network is defined as a virtual private cloud (VPC). External addresses are accessible from the Internet.
- In addition to specifying IP addresses, you will often need to define firewall rules to control access to sub networks and VMs in your VPC.
- For example: you may have a database server that you want to restrict access to so that only an application server can query the database.
- A firewall rule can be configured to limit inbound and outbound traffic to the IP address of the application server or load balancer in front of the application cluster.

Specialized Services

Most public cloud providers offer specialized services that can be used as building blocks of applications or as part of a workflow for processing data:

- They are serverless; you do not need to configure servers or clusters.
- They provide a specific function, such as translating text or analyzing images.
- They provide an application programming interface (API) to access the functionality of the service.
- AutoML, a machine learning service
- Cloud Natural Language, a service for analyzing text
- Cloud Vision for analyzing images
- Cloud Inference API, a service for computing correlations over time-series data.

Specialized services encapsulate advanced computing capabilities and make them accessible to developers who are not experts in domains, such as natural language processing and machine learning.



Cloud Computing vs. Data Center Computing

- ❖ Rent Instead of Own Resources
- ❖ Pay-as-You-Go-for-What-You-Use Model
- ❖ Elastic Resource Allocation
- ❖ Specialized Services

Computing Components of GCP

Google Cloud Platform is a suite of cloud computing services that includes compute, storage, and networking services designed to meet the needs of a wide range of cloud computing customers.

Most of the services can be grouped into several core categories:-

- Computing resources
- Storage resources
- Databases
- Networking services
- Identity management and security
- Development tools
- Management tools
- Specialized services

Computing Resources

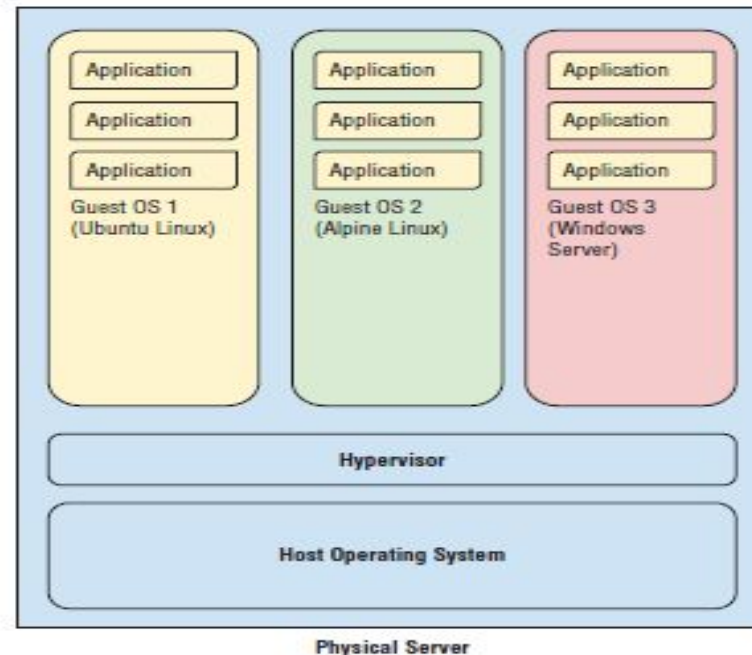
- Public cloud services provide a range computing services options. At one end of the spectrum, customers can create and manage VMs themselves. This model gives the cloud user the greatest control of all the computing services. Users can choose the operating system to run, which packages to install, and when to back up and perform other maintenance operations. This type of computing service is typically referred to as infrastructure as a service(IaaS).
- An alternative model is called platform as a service (PaaS), which provides a runtime environment to execute applications without the need to manage underlying servers, networks, and storage systems.

Different Types of Computing Resource Present in GCP are:-

- ❖ **Compute Engine**
- ❖ **Kuberentes Engine**
- ❖ **App Engine**
- ❖ **Cloud Functions**

Compute Engine

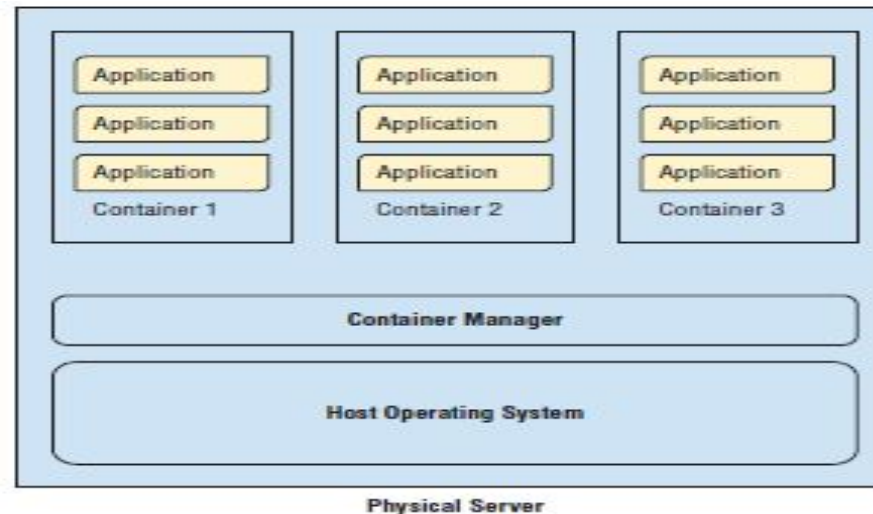
Compute Engine is a service that allows users to create VMs, attach persistent storage to those VMs, and make use of other GCP services, such as Cloud Storage.



VMs run within a low-level service called a *hypervisor*. GCP uses a security hardened version of the KVM hypervisor. KVM stands for Kernel Virtual Machine and provides virtualization on Linux systems running on x86 hardware.

Kubernetes Engine

Kubernetes Engine is designed to allow users to easily run containerized applications on a cluster of servers. Containers are often compared to VMs because they are each used for isolating computing processing and resources. Containers take a different approach than VMs for isolating computing processes.



Kubernetes Engine is a GCP product that allows users to describe the compute, storage, and memory resources they'd like to run their services. Kubernetes Engine then provisions the underlying resources. It's easy to add and remove resources from a Kubernetes cluster using a command-line interface or a graphical user interface.

App Engine

App Engine is GCP's compute PaaS offering. With App Engine, developers and application administrators don't need to concern themselves with configuring VMs or specifying Kubernetes clusters. Instead, developers create applications in a popular programming language such as Java, Go, Python, or Node.js and deploy that code to a serverless application environment.

- App Engine is well suited for web and mobile backend applications.
- **App Engine is available in two types:-**
- ◆ **Standard:** standard environment is well suited to applications that are written in one of the supported languages and do not need operating system packages or other compiled software that would have to be installed along with the application code.
- **Flexible:** flexible environment works well in cases where you have application code but also need libraries or other third-party software installed. As the name implies, the flexible environment gives you more options, including the ability to work with background processes and write to local disk.

Cloud functions

Google Cloud Functions is a lightweight computing option that is well suited to event-driven processing. Cloud Functions runs code in response to an event, like a file being uploaded to Cloud Storage or a message being written to a message queue.

- The code that executes in the Cloud Functions environment must be short-running—this computing service is not designed to execute long-running code. If you need to support long-running applications or jobs, consider Compute Engine, Kubernetes Engine, or App Engine.
- Cloud Functions is often used to call other services, such as third-party application programming interfaces (APIs) or other GCP services, like a natural language translation service.
- Cloud Functions is a serverless product. Users only need to supply code; they do not need to configure VMs or create containers. Cloud Functions will automatically scale as load increases.

Storage Components of GCP

Sometimes an application needs fast read and write times for moderate amounts of data. Other times, a business application may need access to petabytes of archival storage but can tolerate minutes and even hours to retrieve a document. GCP has several storage resources for storing objects and files such as:

Storage Resources

- 1) Cloud Storage**
- 2) Persistent Disk**
- 3) Cloud Storage for Firebase**
- 4) Cloud Filestore**

Databases

- 1) Cloud SQL**
- 2) Cloud Bigtable**
- 3) Cloud Spanner**
- 4) Cloud Datastore**
- 5) Cloud Memorystore**
- 6) Cloud Firestore**

Storage Resources of GCP

1) Cloud Storage: is GCP's object storage system. Objects can be any type of file or binary large object. Objects are organized into buckets, which are analogous to directories in a file system. It is important to remember that Cloud Storage is not a file system. It is a service that receives, stores, and retrieves files or objects from a distributed storage system. Cloud Storage is useful for storing objects that are treated as single units of data.

2) Persistent Disk: Persistent disks are storage service that are attached to VMs in Compute Engine or Kubernetes Engine. Persistent disks provide block storage on solid-state drives (SSDs) and hard disk drives (HDDs). SSDs are often used for low-latency applications where persistent disk performance is important.

3) Cloud Storage for Firebase: Mobile app developers may find Cloud Storage for Firebase to be the best combination of cloud object storage and the ability to support uploads and downloads from mobile devices with sometimes unreliable network connections.

The Cloud Storage for Firebase API is designed to provide secure transmission as well as robust recovery mechanisms to handle potentially problematic network quality.



Storage Resources of GCP

4) Cloud Filestore: Sometimes, developers need to have access to a file system housed on network-attached storage. For these use cases, the Cloud Filestore service provides a shared file system for use with Compute Engine and Kubernetes Engine.

Filestore can provide high numbers of input-output operations per second (IOPS) as well as variable storage capacity. File system administrators can configure Cloud Filestore to meet their specific IOPS and capacity requirements.

Storage Databases of GCP

GCP provides several database options such as:

- Some are relational databases
- some are NoSQL databases
- Some are serverless and others require users to manage clusters of servers.
- Some provide support for atomic transactions, and others are better suited for applications with less stringent consistency and transaction requirements

GCP users must understand their application requirements before choosing a service, and this is especially important when choosing a database, which often provides core storage services in the application stack.

1) Cloud SQL: Cloud SQL is GCP's managed relational database service that allows users to set up MySQL or PostgreSQL databases on VMs without having to attend to database administration tasks, such as backing up databases or patching database software.

Cloud SQL is available in a number of configurations such as:

- ❖ First-generation MySQL databases use MySQL 5.5 or 5.6
- ❖ Second-generation MySQL databases use MySQL 5.6 or 5.7
- ❖ PostgreSQL 9.6 runs on the second-generation platform

Storage Databases of GCP

2) Cloud Bigtable: Cloud Bigtable is designed for petabyte-scale applications that can manage up to billions of rows and thousands of columns. It is based on a NoSQL model known as a *wide-column data model*, and unlike Cloud SQL that supports relational databases. Bigtable is suited for applications that require low-latency write and read operations. It is designed to support millions of operations per second.

3) Cloud Spanner: Cloud Spanner is Google's globally distributed relational database that combines the key benefits of relational databases, such as strong consistency and transactions, with the ability to scale horizontally like a NoSQL database. Spanner is a high availability database with a 99.999 percent availability Service Level Agreements (SLA), making it a good option for enterprise applications that demand scalable, highly available relational database services.

4) Cloud Datastore: Cloud Datastore is a NoSQL document database. This kind of database uses the concept of a document, or collection of key-value pairs, as the basic building block. Cloud Datastore is accessed via a REST API that can be used from applications running in Compute Engine, Kubernetes Engine, or App Engine. This database will scale automatically based on load.

Cloud Datastore is well suited to applications that demand high scalability and structured data and do not always need strong consistency when reading data.

Storage Databases of GCP

5) Cloud Memorystore: Cloud Memorystore is an in-memory cache service. Other databases offered in GCP are designed to store large volumes of data and support complex queries, but Cloud Memorystore is a managed Redis service for caching frequently used data in memory. Caches like this are used to reduce the time needed to read data into an application. Cloud Memorystore is designed to provide submillisecond access to data.

6) Cloud Firestore: Cloud Firestore is another GCP-managed NoSQL database service designed as a backend for highly scalable web and mobile applications. A distinguishing feature of Cloud Firestore is its client libraries that provide offline support, synchronization, and other features for managing data across mobile devices, IoT devices, and backend data stores.

For example: applications on mobile devices can be updated in real time as data in the backend changes.

Network Components of GCP

Google Cloud Platform provides a number of networking services designed to allow users to configure virtual networks within Google's global network infrastructure, link on-premise data centres to Google's network, optimize content delivery, and protect your cloud resources using network security services.

1) Virtual Private Cloud: When an enterprise operates its own data center, it controls what is physically located in that data center and connected to its network. Its infrastructure is physically isolated from those of other organizations running in other data centers.

When an organization moves to a public cloud, it shares infrastructure with other customers of that public cloud. Although multiple enterprises will use the same cloud infrastructure, each enterprise can logically isolate its cloud resources by creating a virtual private cloud (VPC).

VPC can span the globe without relying on the public Internet. Traffic from any server on a VPC can be securely routed through the Google global network to any other point on that network.

2) Cloud Load Balancing: Google provides global load balancing to distribute workloads across your cloud infrastructure. Cloud Load Balancing can distribute the workload within and across regions, adapt to failed or degraded servers, and autoscale your compute resources to accommodate changes in workload.

Cloud Load Balancing is a software service that can load-balance HTTP, HTTPS, TCP/SSL, and UDP traffic.

Network Components of GCP

3) Cloud Armor: Cloud Armor is a Google network security service that builds on the Global HTTP(s) Load Balancing service. Cloud Armor features include the following:

- ❖ Ability to allow or restrict access based on IP address
- ❖ Predefined rules to counter cross-site scripting attacks
- ❖ Ability to counter SQL injection attacks
- ❖ Ability to define rules at both level 3 (network) and level 7 (application)
- ❖ Allows and restricts access based on the geolocation of incoming traffic.

4) Cloud CDN: With content delivery networks (CDNs), users anywhere can request content from systems distributed in various regions. CDNs enable low-latency response to these requests by caching content on a set of endpoints across the globe. Google currently has more than 90 CDN endpoints that are managed as a global resource, so there is no need to maintain region-specific configurations. CDNs are especially important for sites with large amounts of static content and a global audience.

5) Cloud Interconnect: Cloud Interconnect is a set of GCP services for connecting your existing networks to the Google network.

Cloud Interconnect offers two types of connections:- **“interconnects”** Interconnect with direct access to networks uses the Address Allocation for Private Internets standard (RFC 1918) to connect to devices in your VPC. **“peering”** if an organization cannot achieve a direct interconnect with a Google facility, it could use Partner Interconnect.



Network Components of GCP

6) Cloud DNS: Cloud DNS is a domain name service provided in GCP. Cloud DNS is a high availability, low-latency service for mapping from domain names, such as example.com, to IP addresses, such as 74.120.28.18.

Cloud DNS is designed to automatically scale so customers can have thousands and millions of addresses without concern for scaling the underlying infrastructure. Cloud DNS also provides for private zones that allow you to create custom names for your VMs if you need those.

Identity Management

GCP's Cloud Identity and Access Management (IAM) service enables customers to define fine-grained access controls on resources in the cloud. IAM uses the concepts of users, roles, and privileges.

- Identities are abstractions about users of services, such as a human user. After an identity is authenticated by logging in or some other mechanism, the authenticated user can access resources and perform operations based on the privileges granted to that identity.
- For example, a user may have the privilege to create a bucket in Cloud Storage or delete a VM running in Compute Engine.
- Users often need similar sets of permissions. Someone who has the ability to create a VM will likely want to be able to modify or delete those VMs. Groups of related permissions can be bundled into roles. Roles are sets of permissions that can be assigned to an identity.

Development Tools

Google Cloud Platform is an excellent choice for developers and software engineers because of the easy access to infrastructure and data management services, but also for the tools it supports.

- Cloud SDK is a command-line interface for managing GCP resources, including VMs, disk storage, network firewalls, and virtually any other resource you might deploy in GCP. In addition to a command-line interface, Cloud SDK has client libraries for Java, Python, Node.js, Ruby, GO, .NET, and PHP.

Google has also developed plug-ins to make it easy to work with popular development tools. These include the following:

- ❖ Cloud Tools for IntelliJ
- ❖ Cloud Tools for PowerShell
- ❖ Cloud Tools for Visual Studio
- ❖ Cloud Tools for Eclipse
- ❖ App Engine Gradle Plugin
- ❖ App Engine Maven Plugin

Management Tools

The following are some of the most important tools in the management tools category:

- **Stackdriver:** This is a service that collects metrics, logs, and event data from applications and infrastructure and integrates the data so DevOps engineers can monitor, assess, and diagnose operational problems.
- **Monitoring:** This extends the capabilities of Stackdriver by collecting performance data from GCP, AWS resources, and application instrumentation, including popular open source systems like NGINX, Cassandra, and Elasticsearch.
- **Logging:** This service enables users to store and analyze and alert on log data from both GCP and AWS logs.
- **Error Reporting:** This aggregates application crash information for display in a centralized interface.
- **Trace:** This is a distributed tracing service that captures latency data about an application to help identify performance problem areas.
- **Debugger:** This enables developers to inspect the state of executing code, inject commands, and view call stack variables.
- **Profiler:** This is used to collect CPU and memory utilization information across the call hierarchy of an application. Profiler uses statistical sampling to minimize the impact of profiling on application performance.

The combination of management tools provides insights into applications as they run in production, enabling more effective monitoring and analysis of operational systems.

Specialized Services of GCP

In addition to IaaS and PaaS offerings, GCP has specialized services for APIs, data analytics, and machine learning.

- **Apigee API Platform:** The Apigee API platform is a management service for GCP customers providing API access to their applications. The Apigee platform allows developers to deploy, monitor, and secure their APIs. It also generates API proxies based on the Open API Specification.

- **Data Analytics:** GCP has a number of services designed for analyzing big data in batch and streaming modes. Some of the most important tools in this set of services include the following:
 - 1) BigQuery - petabyte-scale analytics database service for data warehousing
 - 2) Cloud Dataflow - a framework for defining batch and stream processing pipelines
 - 3) Cloud Dataproc - a managed Hadoop and Spark service
 - 4) Cloud Dataprep – this service allows analysts to explore and prepare data for analysis

- **AI and Machine Learning:** Google is a leader in AI and machine learning, so it is no surprise that GCP includes several AI services. Specialized services in this area include the following:
 - 1) Cloud AutoML
 - 2) Cloud Machine Learning Engine
 - 3) Cloud Vision
 - 4) Cloud Natural Language Processing

How GCP Organizes Projects and Accounts

When you use GCP, you probably launch virtual machines or clusters, maybe create buckets to store objects, and make use of serverless computing services such as App Engine and Cloud Functions. The list of resources you use can grow quickly.

- GCP provides a way to group resources and manage them as a single unit. This is called the resource hierarchy. The access to resources in the resource hierarchy is controlled by a set of policies that you can define.
- The central abstraction for managing GCP resources is the resource hierarchy. It consists of three levels:
 - 1) **Organization**
 - 2) **Folder**
 - 3) **Project**

1) Organization: An organization is the root of the resource hierarchy and typically corresponds to a company or organization.

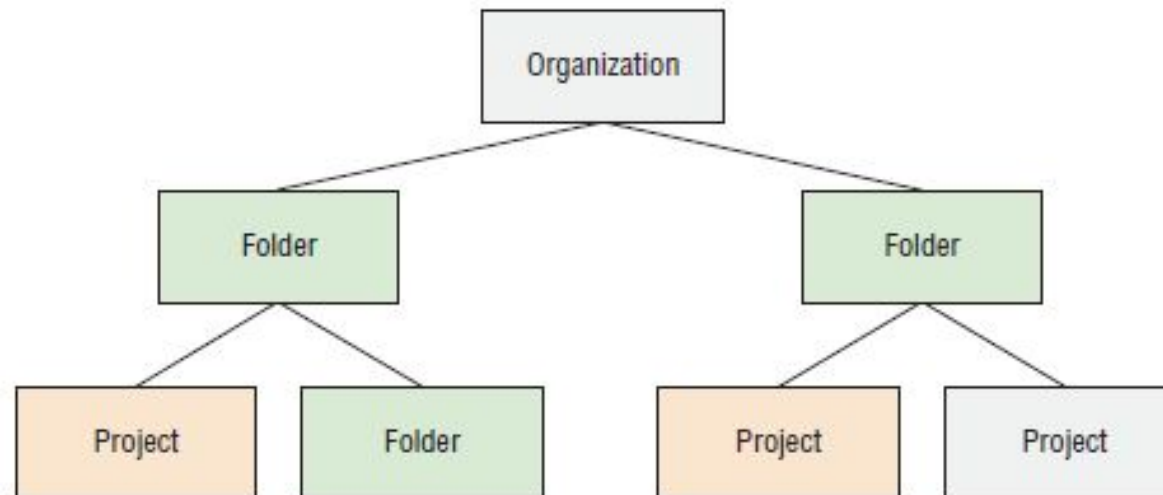
G Suite is Google's office productivity suite, which includes Gmail, Docs, Drive, Calendar, and other services. Cloud identities have super admins, and those super admins assign the role of Organization Administrator Identity and Access Management (IAM) role to users who manage the organization.

How GCP Organizes Projects and Accounts

The users with the Organization Administrator IAM role are responsible for the following:

- Defining the structure of the resource hierarchy
- Defining identity access management policies over the resource hierarchy
- Delegating other management roles to other users.

2) Folder: Folders are the building blocks of multilayer organizational hierarchies. Organizations contain folders. Folders can contain other folders or projects. A single folder may contain both folders and projects.





How GCP Organizes Projects and Accounts

3) Projects: Projects are in some ways the most important part of the hierarchy. It is in projects that we create resources, use GCP services, manage permissions, and manage billing options.

- The first step in working with a project is to create one. Anyone with the resource manager projects create IAM permission can create a project. By default, when an organization is created, every user in the domain is granted that permission.
- Your organization will have a quota of projects it can create. The quota can vary between organizations. Google makes decisions about project quotas based on typical use, the customer's usage history, and other factors. After you have created your resource hierarchy, you can define policies that govern it.

Organization Policies

GCP provides an Organization Policy Service. This service controls access to an organization's resources. The Organization Policy Service lets you specify limits on the ways resources can be used.

Constraints on Resources: Constraints are restrictions on services. GCP has list constraints and Boolean constraints. The following are some types of list constraints:

- Allow a specific set of values
- Deny a specific set of values
- Deny a value and all its child values
- Allow all allowed values
- Deny all values

Boolean constraints evaluate to true or false and determine whether the constraint is applied or not.

Policy Evaluation: Organizations may have standing policies to protect data and resources in the cloud. For example, there may be rules dictating who in the organization can enable a service API or create a service account. Your InfoSec department may require that all VMs disable serial port access.

Roles and Identities

In addition to managing resources, as a cloud engineer you will have to manage access to those resources. This is done with the use of roles and identities.

Roles in GCP: A role is a collection of permissions. Roles are granted to users by binding a user to a role. There are three types of roles in Google Cloud Platform:

- Primitive roles
- Predefined roles
- Custom roles

1) Primitive roles: include Owner, Editor, and Viewer. These are basic privileges that can be applied to most resources. It is a best practice to use predefined roles instead of primitive roles when possible.

2) Predefined roles: provide granular access to resources in GCP, and they are specific to GCP products.

3) Custom roles: allow cloud administrators to create and administer their own roles. Custom roles are assembled using permissions defined in IAM.

Once you have determined which roles you want to provide to users, you can assign roles to users through the IAM console. It is important to know that permissions cannot be assigned to users.

Billing

Using resources such as VMs, object storage, and specialized services usually incurs charges. The GCP Billing API provides a way for you to manage how you pay for resources used.

Billing accounts: store information about how to pay charges for resources used. A billing account is associated with one or more projects. All projects must have a billing account unless they use only free services. Billing accounts can follow a similar structure to the resource hierarchy. There are two types of billing accounts:

1) Self-Serve

2) Invoiced

“**Self-serve accounts**” are paid by credit card or direct debit from a bank account. The costs are charged automatically.

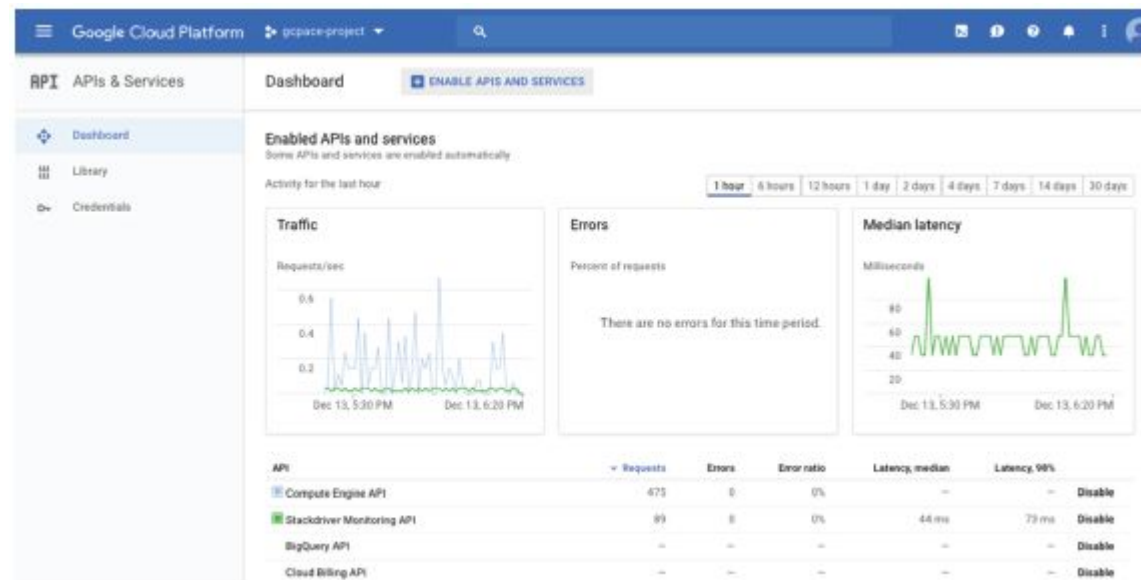
The other type is an “**invoiced billing account**”, in which bills or invoices are sent to customers. This type of account is commonly used by enterprises and other large customers.

Billing Budgets and Alerts: The GCP Billing service includes an option for defining a budget and setting billing alerts. You can navigate to the budget form from the main console menu, selecting Billing and then Budgets & alerts.

Enabling APIs

GCP uses APIs to make services programmatically accessible. For example, when you use a form to create a VM or a Cloud Storage bucket, behind the scenes, API functions are executed to create the VM or bucket. All GCP services have APIs associated with them. Most, however, are not enabled by default in a project.

- To enable service APIs, you can select APIs & Services from the main console menu. This will display a dashboard.



If you click the Enable APIs and Services link, you will see a list of services that you can enable, as shown in above Figure.